Hairball

Manual . free and open source (AGPL-3.0-or-later) . github.com/SpeakingColors/hairball . by Studio 87

Hairball takes sound that already exists and freezes it, smears it, stutters it and turns it into risers. It is not a granular synth built from scratch: it is a transformer and destroyer for audio. You use it in two ways. As an instrument (plugin "Hairball"), you load an audio file and play it with MIDI keys. As an effect (plugin "Hairball FX"), the audio of the track passes through it; you can freeze what it just heard, hold the playing position for a stutter, run a riser, or capture the buffer as a small sample. Both plugins come from the same code and have the same window and the same controls. Hairball FX starts in FX mode and listens to the track; Hairball starts in Instrument mode and does not take an audio input by default. This is a test build: it is not tuned by ear yet, and some numbers (for example the Dance settings) are approximate.

Quick start

• Instrument: drop a WAV, AIFF, FLAC or OGG file on the main display (or click Load audio), then play the piano or your MIDI keyboard.
• FX: put Hairball FX on an audio track and play. Press Freeze to loop what it just heard, or Hold for a stutter.
• Capture turns the buffer into a small sample that you can play with the keys (it switches to Instrument).
• Riser: switch Riser on in the Riser tab and press the Riser button; grains shrink, get closer and fuse into a pitched tone while pitch and filter climb.
• Thicken tab: width, EQ, filter, Drive, Transistor, Fuzz and Dance.
• Key to: Percent, Step and Transient use the keys from C5 up (green marks on the piano) to jump to a place in the sound.
• Save your own sounds with Save. They stay on your computer when you uninstall unless you choose to delete them.

Interface tour

Pink numbers mark controls, teal letters mark displays. Every number is explained in the tables on the next pages. The window has three tabs; one picture per tab follows. The pictures come from the interactive test page of the plugin window; the compiled plugin may differ by a few pixels of layout.

Grains page



Riser page



Thicken page



Signal flow

Instrument: loaded sound (or captured buffer) > grains (Attack, Hold len, Gr. spacing, Wave spacing, Pan, LFO, transients) > Pitch and Amp envelope per voice (up to 8 voices) > Thicken (Width, Bass mono, 3 band EQ) > Filter > Colour (Drive, Transistor, Fuzz) > Dance > Dry/wet and Limiter > Output. FX: the track audio is kept in a ring buffer of up to 8 seconds; Freeze loops it, Lag keeps the grains just behind the live audio; the same chain follows.

Header bar

1	**Previous preset**	Previous preset
2	**Preset menu**	Preset list: factory presets first, then your own.
3	**Next preset**	Next preset
4	**Save**	Save the current sound. Overwrite your own preset or save as a new one.
5	**Rename**	Rename your preset.
6	**Delete**	Delete your preset. Factory presets cannot be deleted.
7	**Export**	Show the preset as text you can copy.
8	**Import**	Paste preset text. It is checked before it is used.
9	**Audio (mockup page only)**	Audio on or off. Browsers need one click before they play sound.
10	**Guide**	Opens the short guide: how to use Instrument and FX, Freeze, Hold, Capture, Riser and Thicken, the Key to keys, and how to see a hint.
11	**Settings**	Menu: info bubbles on or off, power saving, show grains on the display, reset the window size to 100%, open the GitHub page.

12	**About & Support**	Studio 87, version, licence (AGPL-3.0-or-later), where your saved sounds live, and the Support on Ko-fi button. Nothing is sent anywhere; the button only opens a link when you click it.
13	**Window size**	Window size, percent of the default. Drag any edge or corner to resize; double click the wood to go back to 100%.

Source, display and keys (display A, B)



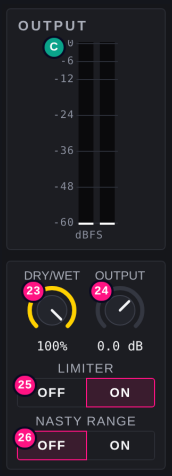
Display A: Main display. Instrument: the loaded sound as a waveform. FX: the audio the track sent, newest on the right; it turns blue while frozen. Dashed green line: the Start position (Instrument: click anywhere on the display to set it). Orange ticks at the top: transient marks when Transients is Regions or Detect. White marks: where grains are reading (switch off with Show grains on the display in Settings). Pink lines: where the voices are playing now. Bottom right: length of the sound, or the buffer length in seconds. Drop an audio file on it to load it.

Display B: Piano keys (36 keys, with Octave - and +). Click a key or use your MIDI keyboard. Computer keys: Z S X D C V G B H N J M and Q 2 W 3 E R 5 T 6 Y 7 U I. When Key to is Percent, Step or Transient, the keys from C5 up are marked green: they jump to a place in the sound instead of changing the pitch.

14	**Mode**	Instrument plays the loaded sample with MIDI. FX works on the audio passing through the track. (Instrument / FX)
15	**Audio source**	Instrument: the sound you play with the keys. FX: the audio that passes through the track. In the plugin FX listens to the track; here you pick a loop or drop a file.
16	**Freeze**	FX: stop recording and loop what is in the buffer. Grains then play only the frozen audio.
17	**Hold**	Holds the playing position still. Click to latch, or press and hold for a stutter.
18	**Play**	Instrument: plays the sound at its root key and keeps playing until you press again. You can also click the piano keys. FX: the loop plays by itself.

19	Riser	Starts the riser: grains shrink, get closer and fuse into a pitched tone while pitch and filter climb.
20	Capture	FX: record what the buffer holds into Hairball as a small sample and switch to Instrument to play it.
21	Capture	How many seconds of the incoming audio Hairball keeps ready to freeze (FX mode). (0.50 s to 8.00 s, default 4.00 s)
22	Lag	FX mode, not frozen: how far behind the live audio the grains read. (0.0 ms to 2.00 s, default 120 ms)

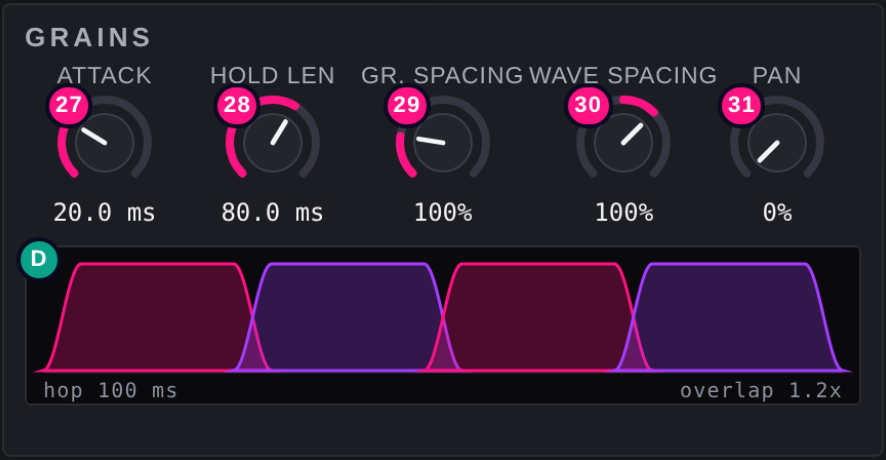
Output (display C)



Display C: Output meter, in dBFS. The translucent bar is the peak, the solid inner bar the level (RMS), and a line holds the peak for 1.5 seconds. Neon blue is safe, purple is a warning, pink is clipping.

23	Dry/wet	Blend between the untouched input and Hairball. At 0 the input passes bit for bit. (0% to 100%, default 100%)
24	Output	Output level. (-24.0 dB to +12.0 dB, default 0.0 dB)
25	Limiter	Safety limiter at the output, never above 0 dBFS. (Off / On)
26	Nasty range	Lets Drive, Transistor and Fuzz go beyond 100 percent. Off by default. (Off / On)

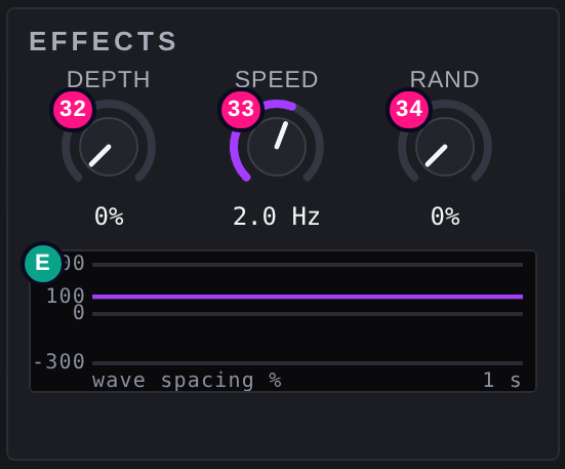
Grains (display D) [Grains page]



Display D: Grain shape. Four grains one after another: pink and violet shapes. Each grain fades in over Attack, holds for Hold len, and fades out over Attack. The distance between grain starts is the hop (shown bottom left in ms); the overlap (shown bottom right) is how many grains cover each other.

27	Attack	Fade in and fade out of each grain. Added to the grain length on both ends. (0.0 ms to 250 ms, default 20.0 ms)
28	Hold len	Length each grain plays before the next one takes over. (2.0 ms to 800 ms, default 80.0 ms)
29	Gr. spacing	Gap between grains. 100 percent places them end to end with a clean crossfade. Lower means more overlap and a thicker sound, higher means gaps. (25% to 400%, default 100%)
30	Wave spacing	How fast the playing position moves through the sound. 100 is normal, 0 freezes, negative plays the grains backwards in order, 300 is three times faster. (-300% to 300%, default 100%)
31	Pan	Odd grains go left, even grains go right. 0 is off. (0% to 100%, default 0%)

Effects (display E) [Grains page]

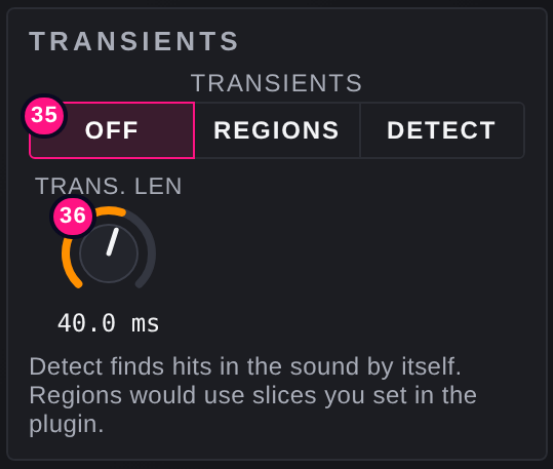


Display E: LFO and wave spacing, over one second. Vertical: Wave spacing in percent (-300 to +300); the violet line is Wave spacing plus the LFO (Depth, Speed) moving around it. Flat when Depth is 0.

32	Depth	Size of the LFO that moves Wave spacing. 0 is off. (0% to 300%, default 0%)
33	Speed	Speed of the LFO on Wave spacing. (0.1 Hz to 30.0 Hz, default 2.0 Hz)

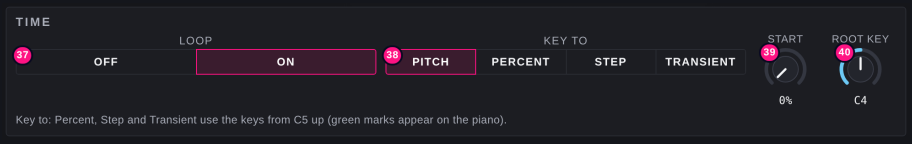
34	Rand	Randomises where each grain starts, so the order is shuffled. 0 is off. (0% to 100%, default 0%)

Transients [Grains page]



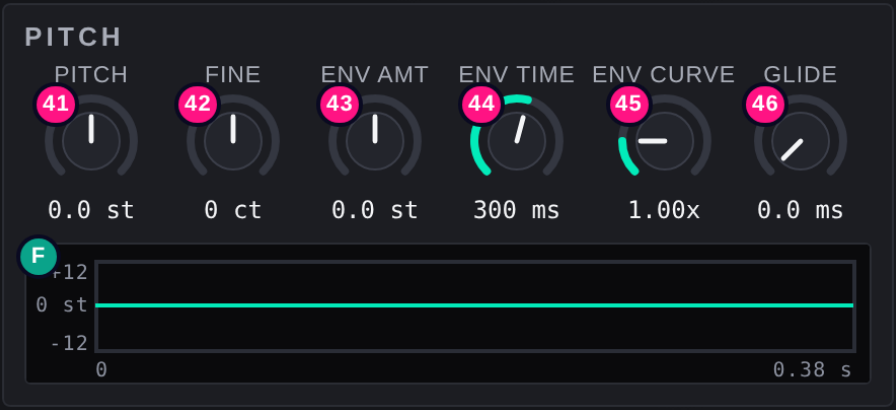
35	Transients	Grains that cross a hit snap to it and take the Transient length. Off, Regions (slices you provide) or Detect (automatic). (Off / Regions / Detect)
36	Trans. len	Length of grains that start on a transient. (2.0 ms to 400 ms, default 40.0 ms)

Time [Grains page]



37	Loop	Loop the sound when the position reaches the end. (Off / On)
38	Key to	What the MIDI keys do: Pitch, Percent (C5 to C7 pick the start position), Step (C5 and up offset the start by steps), Transient (C5 and up pick a slice). (Pitch / Percent / Step / Transient)
39	Start	Where in the sound playing starts. (0% to 100%, default 0%)
40	Root key	The MIDI note that plays the sound at its original pitch. (C1 to C7, default C4)

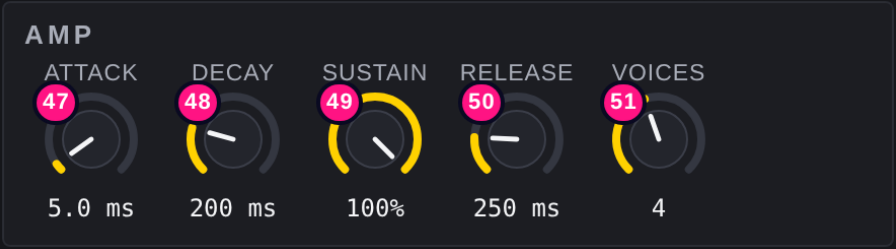
Pitch (display F) [Riser page]



Display F: Pitch envelope. Horizontal: time from the note start. Vertical: semitones (the scale grows with Env amt). The green line starts at Env amt, goes to 0 st over Env time and is bent by Env curve.

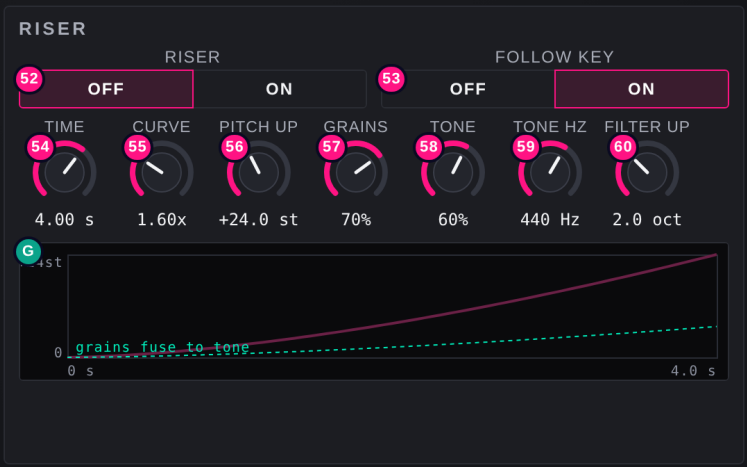
41	**Pitch**	Pitch in semitones. Grains are read faster or slower; the position still moves at Wave spacing. (-48.0 st to +48.0 st, default 0.0 st)
42	**Fine**	Fine tune in cents. (-100 ct to +100 ct, default 0 ct)
43	**Env amt**	Pitch envelope. Starts this many semitones away from the pitch and glides back to it. (-48.0 st to +48.0 st, default 0.0 st)
44	**Env time**	How long the pitch envelope takes to return to the set pitch. (5.0 ms to 8.00 s, default 300 ms)
45	**Env curve**	Shape of the pitch envelope. Below 1 snaps back fast, above 1 lingers away from the pitch. (0.20x to 5.00x, default 1.00x)
46	**Glide**	Instrument: time to slide between notes. (0.0 ms to 2.00 s, default 0.0 ms)

Amp [Riser page]



47	**Attack**	Instrument: note fade in. (0.0 ms to 4.00 s, default 5.0 ms)
48	**Decay**	Instrument: fall to the sustain level. (0.0 ms to 4.00 s, default 200 ms)
49	**Sustain**	Instrument: level while the key is held. (0% to 100%, default 100%)
50	**Release**	Instrument: note fade out. (0.0 ms to 8.00 s, default 250 ms)
51	**Voices**	Instrument: how many notes can play at once. Fewer voices use less CPU. (1 to 8, default 4)

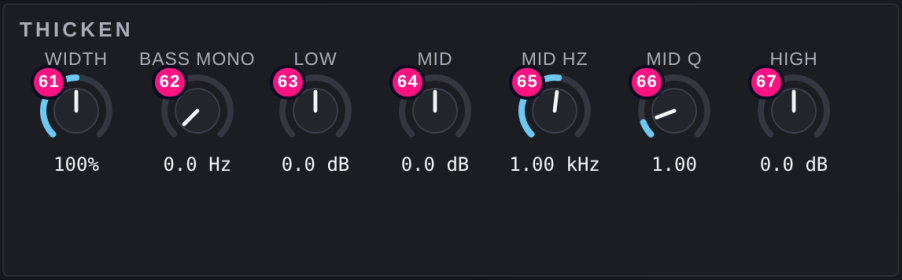
Riser (display G) [Riser page]



Display G: Riser. Horizontal: time, 0 s to the Time setting. Vertical: the pitch climb, 0 to Pitch up. The pink line is the riser progress (bent by Curve; dim when the Riser is off). The dashed green line is the tone share (Tone). When a riser is running, a white dot shows where it is.

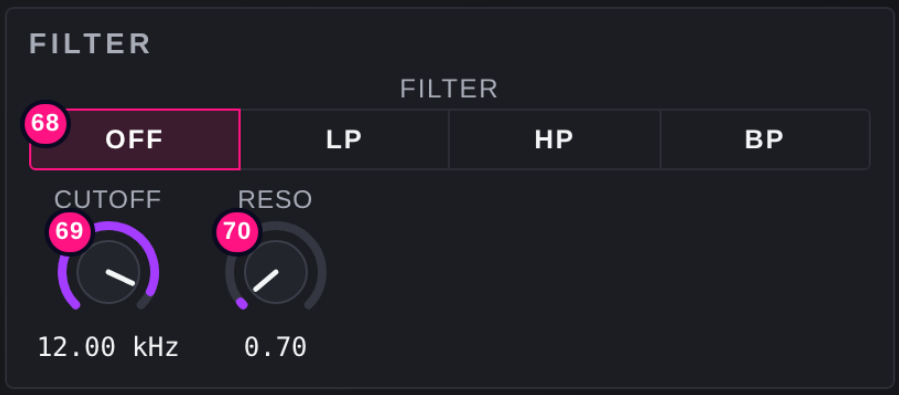
52	**Riser**	Starts the riser. In Instrument mode every note starts it. Turn it off and it eases back. (Off / On)
53	**Follow key**	Instrument: the riser tone follows the key you play. (Off / On)
54	**Time**	How long the riser takes to reach its end. (0.10 s to 32.00 s, default 4.00 s)
55	**Curve**	Below 1 jumps up early, above 1 builds slowly and shoots up at the end. (0.20x to 5.00x, default 1.60x)
56	**Pitch up**	How many semitones the riser lifts the pitch. (0.0 st to +60.0 st, default +24.0 st)
57	**Grains**	How far the grains shrink and speed up as the riser builds. (0% to 100%, default 70%)
58	**Tone**	Grains get so close they turn into a pitched tone at the Tone Hz (or the key you play). 0 stays granular. (0% to 100%, default 60%)
59	**Tone Hz**	Pitch the grains settle on at the top of the riser (FX mode). (40.0 Hz to 2.00 kHz, default 440 Hz)
60	**Filter up**	Opens the filter by this many octaves while the riser builds. (0.0 oct to 6.0 oct, default 2.0 oct)

Thicken [Thicken page]



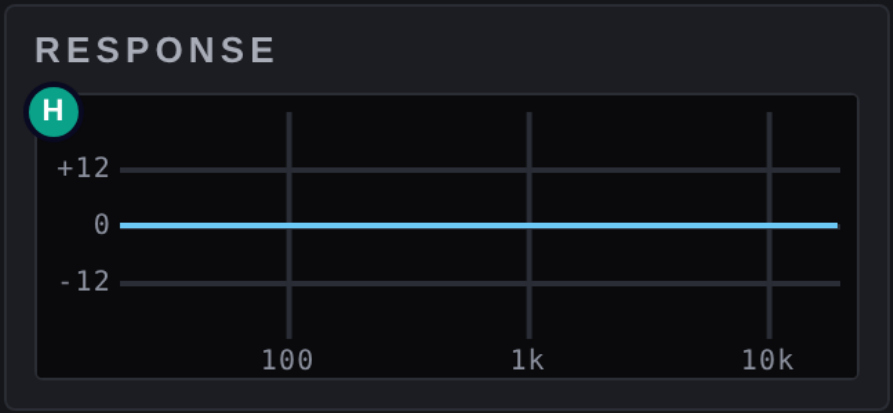
61	Width	Mid/side width. 100 is unchanged, 0 is mono, 200 is double the side signal. (0% to 200%, default 100%)
62	Bass mono	Keeps everything below this frequency in the middle. 0 is off. (0.0 Hz to 400 Hz, default 0.0 Hz)
63	Low	Low shelf at 120 Hz. (-18.0 dB to +18.0 dB, default 0.0 dB)
64	Mid	Bell filter gain. (-18.0 dB to +18.0 dB, default 0.0 dB)
65	Mid Hz	Bell filter frequency. (100 Hz to 8.00 kHz, default 1.00 kHz)
66	Mid Q	Bell filter width. Higher is narrower. (0.30 to 8.00, default 1.00)
67	High	High shelf at 8 kHz. (-18.0 dB to +18.0 dB, default 0.0 dB)

Filter [Thicken page]



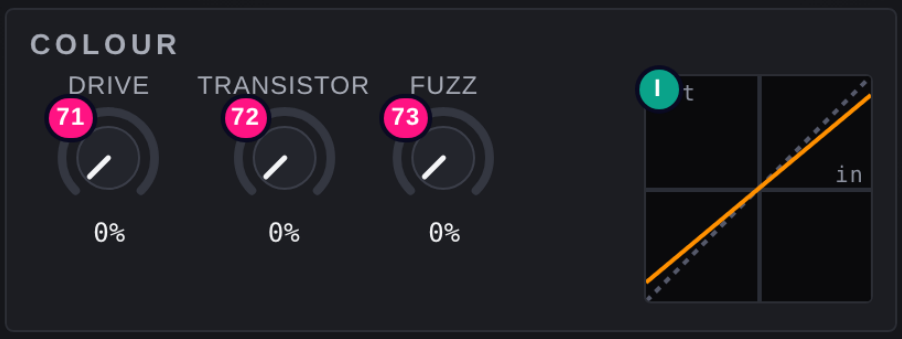
68	Filter	Off, Low pass, High pass or Band pass. (Off / LP / HP / BP)
69	Cutoff	Filter cutoff. The riser can open it by itself. (20.0 Hz to 20.00 kHz, default 12.00 kHz)
70	Reso	Filter resonance. (0.50 to 12.00, default 0.70)

Response (display H) [Thicken page]



Display H: Response. Horizontal: frequency, 20 Hz to 20 kHz (log). Vertical: -24 to +24 dB. The blue line is the sum of Low (shelf at 120 Hz), Mid (bell at Mid Hz), High (shelf at 8 kHz) and the Filter. When the Riser is on, the dashed pink line shows where the filter ends up; while it runs, a dashed white line shows where it is now.

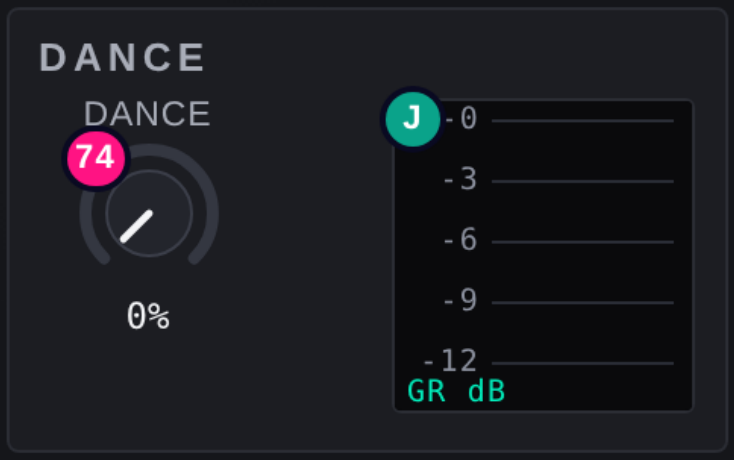
Colour (display I) [Thicken page]



Display I: Transfer curve of the Colour stages (Drive, Transistor, Fuzz). Horizontal: input level (-1 to +1), vertical: output level. The dashed grey diagonal is no change; the orange line is what the stages do to the signal. The curve is drawn after level matching.

71	**Drive**	Symmetric analog clip. Odd harmonics, loud and bright. Above 100 percent needs Nasty range. (0% to 200%, default 0%)
72	**Transistor**	Neve style asymmetric stage. Even harmonics, warm and thick. Above 100 percent needs Nasty range. (0% to 200%, default 0%)
73	**Fuzz**	Lopsided hard fuzz. Above 80 percent it turns nasty on purpose. Above 100 percent needs Nasty range. (0% to 200%, default 0%)

Dance (display J) [Thicken page]



Display J: Dance gain reduction. A green bar grows downward from 0 dB to -12 dB scale while the Dance stage turns the level down; it moves with the sound.

74	Dance	Glue compressor on everything above 150 Hz, bass left alone. Pumps and holds the sound together for dance music. (0% to 100%, default 0%)

Recommendations

• Pad from any sound: Freeze or a long loaded sound, Hold len 300 to 800 ms, Attack 100 to 250 ms, Gr. spacing 50 to 100 percent, then a little Width and Dance.
• Smear: raise Attack so grains overlap, add a slow Depth on the LFO (Effects) for a moving texture.
• Stutter: Hold in FX mode, short Hold len (10 to 40 ms) and Gr. spacing 100 percent.
• Riser: Riser on, Time 4 to 16 s, Pitch up 12 to 36 st, raise Filter up for brightness, and keep the Limiter on.
• If it gets harsh: lower High in Thicken, or turn Drive down before Fuzz.
• Keep the Limiter on while you experiment, the Colour stages can get loud.

Factory presets

Init (Init), Freeze Pad (FX), Cloud Wash (Grains), Reverse Wash (Grains), Riser Tone Sweep (Riser), Riser Up On The Key (Riser), Hold Stutter (FX), Pitch Drop (Pitch), Thick Low Grains (Thicken), Telephone Wobble (FX), Hit Slicer (Rhythm)

Under the hood

• Parameters: 57, all host parameters (automatable). Every control has an info bubble after 1 second (can be switched off in Settings).
• Latency: 0 samples (reported to the host as such). Nothing looks ahead.
• Smoothing: level, mix, width, EQ, filter, Colour and Dance values move over about 10 ms, so a knob change is heard within one audio buffer without clicks.
• Engine: up to 8 voices, up to 24 grains alive per voice, processing in blocks of 32 samples. FX mode keeps up to 8 seconds in a ring buffer. In core tests 8 voices with dense grains, drive, Dance, filter and EQ ran at about 17 times real time (about 6 percent of one core at 48 kHz); the default FX chain at about 220 times real time (0.5 percent of one core). Measured on a test machine, not on yours.
• Limiter: when on, it only touches the wet path (so Dry/wet 0 stays bit exact). It is a soft limiter above about -0.9 dBFS that rounds peaks toward 0 dBFS; it does not look ahead.
• Nasty range: lets Drive, Transistor and Fuzz go from 100 up to 200 percent. Without it they stop at 100 percent.
• Dance: a glue stage that compresses everything above 150 Hz and leaves the bass alone. The amount moves threshold (from -6 to -30 dB) and ratio (from 2:1 to 6:1) together. The numbers are approximate and not tuned by ear.
• Colour: Drive is a symmetric clip, Transistor is asymmetric (even harmonics), Fuzz is lopsided and gets nasty above 80 percent. The stages are level matched, so compare by ear with the knob at zero and at full. The stages are not oversampled yet, and aliasing has not been measured.
• A loaded file is remembered by its location (path), not copied into the session: keep the file where it is, or Hairball shows a message that the sound file was not found. Captured audio is stored inside the session (16 bit, up to 3,000,000 frames).
• Samples are loaded in the interface thread and handed to the audio thread without locks; nothing on the audio thread waits for a file, a lock or the network.
• Presets: factory presets first, then your own. Your own are stored as small text files; Export shows the preset as text you can copy, Import checks pasted text before using it.
• Not tested yet: a real DAW session, the installers on real machines, and whether every host treats the instrument and the FX plugin the way it should. The pictures in this tour come from the interactive test page of the plugin window; the compiled window may differ by a few pixels.

Install, remove and your saved presets

• Windows: run Hairball-Setup.exe. To remove it use Settings, Apps, Hairball. The uninstaller asks whether to also delete your saved presets.

• macOS: open Hairball-macOS.pkg. To remove it run "Uninstall Hairball" from the Applications folder; it asks whether to also delete your saved presets.
• Linux (Flatpak DAW): flatpak install --user the .flatpak file; remove with flatpak uninstall. Linux (native DAW): extract the tarball into ~/.vst3 and delete the Hairball.vst3 and Hairball FX.vst3 folders to remove.
• Your saved presets live in a folder named Studio 87/Hairball/presets in your user application data folder (details in INSTALL.txt).